

Lavkush Gupta

Machine Learning Engineer

Mumbai, India lavkushsg@gmail.com LinkedIn/in/lavkushsg github.com/lavkush

Profile

Machine Learning Engineer (3 YOE) with hands-on experience deploying scalable ML and computer vision systems across edge devices (Jetson Nano), mobile, and cloud (GCP, Azure). Led end-to-end MLOps pipelines, real-time YOLOv8 deployments, and API-first automation systems reducing costs and boosting performance. Patent holder with a strong record in AI prototyping, hackathons, and product-focused ML delivery.

Skills

- Languages:** Python, SQL, JavaScript (for prototyping)
- ML & CV Frameworks:** PyTorch, TensorFlow, Scikit-learn, YOLOv5/v8, Detectron2, ONNX, TensorRT, DeepStream
- NLP & GenAI:** DistilBERT, RASA, NLTK, OpenAI, Gemini
- Deployment & MLOps:** Docker, GitHub Actions, MLflow, FastAPI, Streamlit, Flask, Azure ML Studio, GCP, Kubernetes
- Cloud & Infra:** Azure VM, GCP Compute Engine, AWS (basic), Azure Blob Storage
- Data & Visualization:** Pandas, NumPy, MySQL, PostgreSQL, MongoDB, Power BI
- Other Tools:** Selenium, BeautifulSoup, Git, Next.js, Express.js

Education

- MSc, Computer Science (Data Science Specialization) University of Mumbai** 2019 – 2021 | Mumbai, India
- MSc, Physics, NES Ratnam College of Art Science and Commerce. (Affiliated to University of Mumbai)** 2018 – 2020 | Mumbai, India

Experience

Machine Learning Engineer, SuperZop Feb 2023 – present | Mumbai, India

- Awarded a patent** for a color calibration card and deep learning API used for grain quality analysis in agricultural supply chains.
- Managed scalable cloud infrastructure** on **Azure** and **GCP** (VMs, storage), supporting model retraining and data pipelines.
- Automated real-time reporting** with **SQL + Power BI** dashboards, cutting reporting delays by **80%**.
- Developed a **Flask-based identity verification API** (PAN, Aadhaar, FSSAI), reducing onboarding time by **80%**.
- Optimized segmentation and object detection models** (YOLOv8, TensorRT) for **mobile**, reducing latency by **30%**.
- Built and deployed **end-to-end MLOps pipelines** using **Docker** and **GitHub Actions**, reducing deployment time by **60%** and training costs by **40%**.
- Deployed YOLOv8 and Detectron2 models** on **Jetson Nano (24–28 FPS)**, improving classification accuracy by **35%** and reducing manual quality checks.

Internal Hackathon Project – AI-Powered Traceability Platform

- Delivered a **full-stack internal traceability system** (Next.js + Express + MySQL) integrating **LLMs** and real-time warehouse operations.
- Embedded **live analytics dashboards** inside a **chat interface**, allowing staff to fetch **KPIs and trigger workflows** instantly.
- Orchestrated OpenAI and Gemini LLMs** to convert **natural language** into structured warehouse actions (form-filling, trace logs, API calls).
- Led the project from ideation to deployment**; received **internal award** for best AI-led automation tool during the annual hackathon.

Data Science/Technology Intern, SuperZop Jul 2022 – Jan 2023 | Mumbai

- Built a **color correction model** using **OpenCV** and custom calibration cards, ensuring precise color calibration in CV pipelines.
- Developed a **perspective correction system** for bird’s-eye visualization, improving object localization and measurement accuracy.
- Enhanced object detection pipelines by **optimizing dimension estimation** using color and perspective transformations, improving accuracy by **5%**.

Courses and Certificates

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| - AWS Practical Data Science Specialization | - Machine Learning | - Machine Learning Engineering for Production (MLOps) Specialization |
| - Probability and Statistics: To p or not to p? | - Deep Learning Specialization | |

Languages

- English
- Hindi
- Marathi